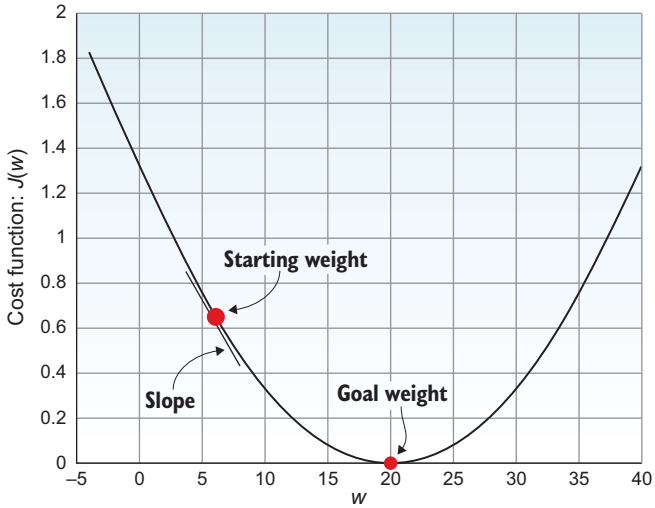

Figure 2.26 The error function with respect to its weight for a single perceptron is a 2D curve.

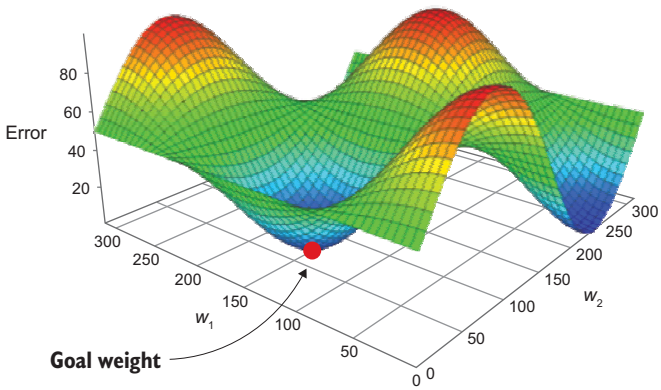

Figure 2.27 Graphing all possible values of two weights gives a 3D error plane.

Since we humans are only equipped to understand a maximum of 3 dimensions, it is impossible for us to visualize error graphs when we have 10 weights, not to mention hundreds or thousands of weight parameters. So, from this point on, we will study the error function using the 2D or 3D plane of the error. In order to optimize the model, our goal is to search this space to find the best weights that will achieve the lowest possible error.

Why do we need an optimization algorithm? Can't we just brute-force through a lot of weight values until we get the minimum error?

Suppose we used a brute-force approach where we just tried a lot of different possible weights (say 1,000 values) and found the weight that produced the minimum error. Could that work? Well, theoretically, yes. This approach might work when we